SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

Signals and Systems: Fourier Series, Transforms, Laplace & Z-Transform

Department: ECE	**Course / Subject:** Signals and Systems (EC401PC)
Semester: Semester 4	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: Continuous & Discrete-time signals, Linear Time-Invariant (LTI) systems, Convolution integrals, Continuous-time Fourier Transform (CTFT), Laplace Transform, and Z-Transform.

Unit 1: Signal Classification & Operations

Continuous vs Discrete-time signals, Energy vs Power signals, Periodic, Even/Odd signals, Unit Step, Impulse, Ramp functions, Time shifting, scaling, reversal.

Unit 2: Linear Time-Invariant (LTI) Systems

LTI system properties (Linearity, Time-invariance, Causality, BIBO Stability), Impulse response, Convolution integral and Convolution sum calculations.

Unit 3: Fourier Series & Fourier Transform

Dirichlet conditions, Trigonometric and Exponential Fourier series, Continuous-Time Fourier Transform (CTFT) properties, Frequency response of LTI systems.

Unit 4: Laplace Transform

Region of Convergence (ROC) properties, Inverse Laplace Transform, System transfer function H(s), Poles and Zeros plotting, Transient and steady-state response.

Unit 5: Z-Transform & Sampling

Z-transform definition and ROC properties, Inverse Z-transform methods (Partial fraction, Power series), Nyquist-Shannon Sampling Theorem, Aliasing mitigation.